

The Aharonov–Bohm Effect

Gauge Potentials, Quantum Phase, and Experimental Evidence

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Abstract. This note gives a self-contained exposition of the Aharonov–Bohm effect for advanced undergraduate students. Starting from the role of electromagnetic potentials in classical electrodynamics, it introduces gauge freedom, minimal coupling, and the gauge covariance of the Schrödinger equation, and derives the Aharonov–Bohm phase for both the time-dependent scalar and the magnetic configurations. It then reviews the experimental evidence, culminating in the 1986 toroidal-magnet experiment of Tonomura and collaborators, in which superconducting shielding confined the magnetic flux completely. The note is expository and does not present original research results.

Keywords: Aharonov–Bohm effect; gauge invariance; vector potential; quantum phase; electron interferometry.

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Chapter 1

Theoretical Foundations

Before discussing the Aharonov–Bohm effect, it is useful to clarify the role played by potentials in classical and quantum physics. We begin with conservative forces and scalar potentials, before extending the discussion to electromagnetism, where the electric and magnetic fields can be expressed in terms of scalar and vector potentials. While in classical physics potentials may appear mainly as convenient mathematical descriptions of the fields, in quantum mechanics they enter directly into the dynamics and the phase of the wavefunction. These ideas provide the conceptual and mathematical foundation needed to understand the Aharonov–Bohm effect.

1.1 Potential energy and potentials

A force is called conservative if the work it performs between two points is independent of the path followed. Equivalently,

$$\oint F \cdot dr = 0.$$

Infinitesimally,

$$\delta W = F \cdot dr.$$

By the work-energy theorem we have that the work performed is equal to the variation of the kinetic energy K ,

$$\delta W = dK.$$

We want to introduce a quantity U by requiring $K + U$ to be conserved. Since the work done by a conservative force depends only on the endpoints, it can be represented by a scalar function,

$$dK + dU = 0.$$

It follows that

$$dU = -dK = -\delta W = -F \cdot dr.$$

Since U is a scalar function of the position, its differential is

$$dU = \nabla U \cdot dr,$$

where ∇ , read as “nabla”, is the gradient operator:

$$\nabla U = \left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z} \right).$$

Comparing the two expressions for dU we obtain:

$$(\nabla U + F) \cdot dr = 0.$$

Since this holds for every infinitesimal displacement dr ,

$$F = -\nabla U.$$

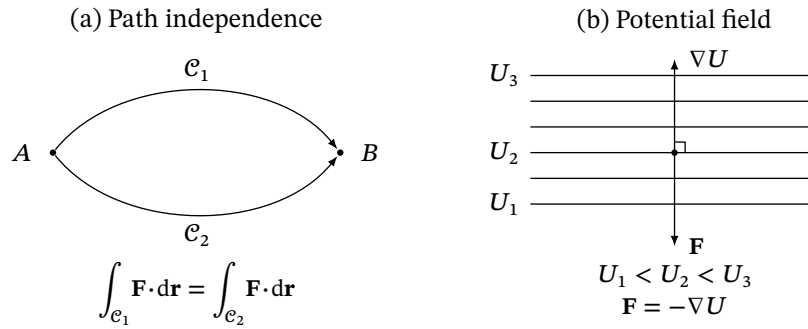


Figure 1.1. Geometric properties of a conservative force. (a) The work between A and B is independent of the path. (b) Horizontal lines are equipotentials: the gradient is perpendicular to them and points toward increasing U , while the force points in the opposite direction.

To visualize the gradient let's consider a body of mass m in a gravitational field. If y denotes its height, its gravitational potential energy is

$$U(y) = mgy.$$

Its gradient is

$$\nabla U = \left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z} \right) = (0, mg, 0).$$

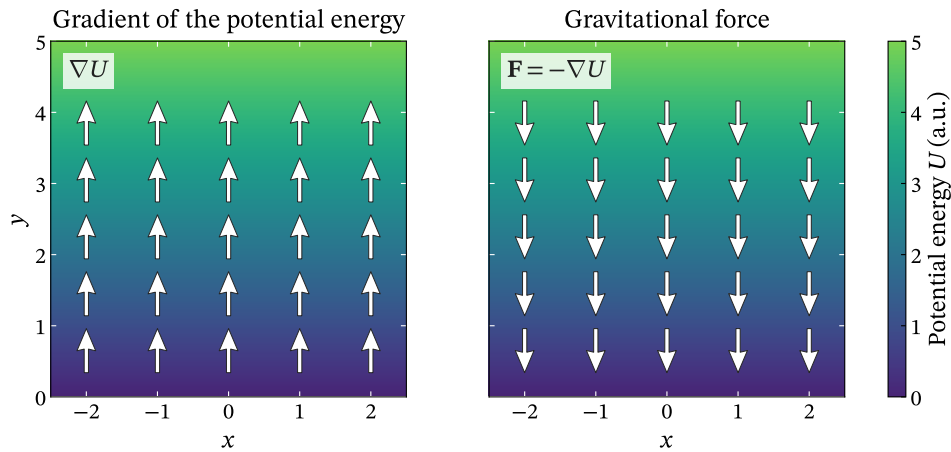


Figure 1.2. The potential-energy gradient and the corresponding gravitational force in a uniform gravitational field.

The gradient, as shown in fig. 1.2, points upward, in the direction in which the potential energy increases most rapidly. The gravitational force is

$$F = -\nabla U = (0, -mg, 0),$$

and therefore points downward, toward the decreasing potential energy.

1.1.1 The potential

The potential energy is not an exclusive property of the field; it depends also on the body used to probe it.

Near the surface of the Earth, for instance,

$$U(y) = mgy.$$

At the same height y , two bodies of different mass have different potential energies. This does not mean that they experience different gravitational fields: the field is the same, and what changes is the amount of matter on which it acts.

We can therefore separate the two contributions by dividing the potential energy by the property of the body that fixes the strength of the interaction; in the gravitational case, its mass:

$$\phi(r) = \frac{U(r)}{m}.$$

The function $\phi(r)$ is called the gravitational potential, and consequently, in the gravitational case,

$$U(r) = m\phi(r).$$

Here ϕ carries the information about the field, while m the information about the probe. Taking the gradient

$$F = -\nabla U = -m\nabla\phi \implies g \equiv \frac{F}{m} = -\nabla\phi,$$

so the potential stands to the field exactly as the potential energy stands to the force.

The same construction applies to any conservative interaction, once the coupling is factored out. For the electrostatic force the role of the mass is played by the electric charge q :

$$V(r) = \frac{U(r)}{q}, \quad E = -\nabla V,$$

where V is the electric potential.

Notice now that U never enters the physics by itself: it enters only through its gradient. This has an immediate consequence. If c is any constant and we replace

$$U(r) \mapsto U(r) + c,$$

then, since differentiation annihilates constants, $\nabla c = 0$ and

$$\nabla(U + c) = \nabla U.$$

The force is unchanged, and so is every trajectory it produces. The same holds for the potential, $\phi \mapsto \phi + c/m$. Two potential energies that differ by a constant are therefore physically indistinguishable: they are not two competing descriptions of which one is correct, but two labels for the same identical physical situation.

1.2 Electromagnetic potentials

We have established that a conservative force that depends on position can be written using a scalar function

$$\mathbf{F} = -\left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z}\right).$$

Now we ask if the same construction can fully describe the electromagnetic force.

The complete electromagnetic force is given by the Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

While the electrostatic contribution depends only on position, the magnetic contribution depends explicitly on the velocity of the charged particle:

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B},$$

and therefore cannot be obtained from the gradient of a potential energy depending only on position. A scalar potential alone is no longer sufficient. A useful consequence of this velocity dependence is that the magnetic force does no work on the particle. Indeed,

$$\mathbf{F}_B \cdot \mathbf{v} = q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = 0,$$

because $\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$. Therefore, a magnetic field may change the direction of the particle's velocity, but not its magnitude or kinetic energy. This means that the magnetic field requires a different kind of potential.

1.2.1 Mathematical interlude: the divergence and curl operators

The divergence of a vector field $\vec{F} = (F_x, F_y, F_z)$ is defined as the dot product:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

It returns a scalar. It measures how much the field locally “flows out of” or “flows into” a point. If the divergence is positive, the point behaves like a source; if it is negative, like a sink; if it is zero, there is no net local outgoing flux.

The curl of $\vec{F}$ is defined as the cross product:

$$\nabla \times \vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right).$$

Notice how the same rule for the cross product between two vectors was used:

$$(a, b, c) \times (d, e, f) = (bf - ce, cd - af, ae - bd),$$

and considering:

$$\vec{F} = (F_x, F_y, F_z), \quad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right).$$

It returns a vector and measures the local tendency of the field to generate a rotation. In other words, the curl tells us whether, in an infinitesimal neighbourhood of a point, the field tends to make a small object placed there rotate.

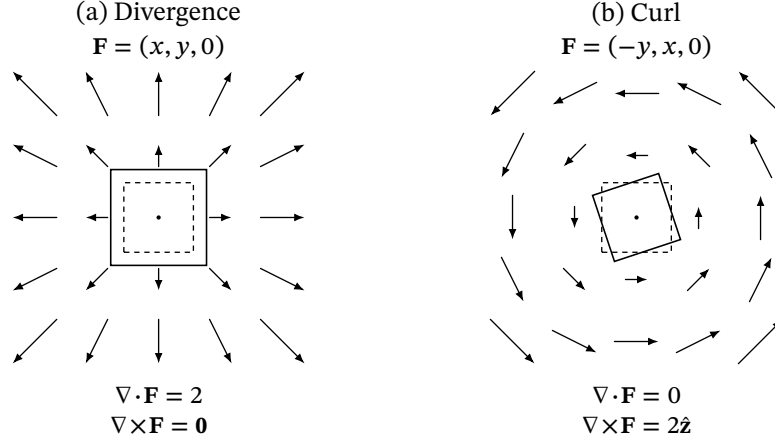


Figure 1.3. Two vector fields in the xy -plane, with x to the right, y upwards and z out of the page. Dashed and solid outlines show a small material element before and after transport by the field. (a) The radial flow expands the element. (b) The rotational flow turns it anticlockwise without changing its area.

Divergence and curl therefore describe two different aspects of a vector field: divergence measures its “radial” behaviour, that is, how much the field locally expands or contracts; curl instead measures its “circular” behaviour, that is, how much the field tends to rotate. This distinction is illustrated in fig. 1.3.

1.2.2 The magnetic vector potential

One of Maxwell’s equations states that the magnetic field is divergence-free:

$$\nabla \cdot \mathbf{B} = 0.$$

In classical electromagnetism, this expresses the absence of isolated magnetic sources.

A standard result of vector calculus states that a sufficiently regular divergence-free vector field can be expressed, at least locally, as the curl of another vector field. We may therefore introduce a vector field $\mathbf{A}(\mathbf{r}, t)$ such that

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

The field $\mathbf{A}$ is called the magnetic vector potential. Unlike the scalar potential introduced previously, $\mathbf{A}$ does not represent a potential energy per unit charge.

To understand its relation to the magnetic field, consider a closed curve C . At each point along the curve, the infinitesimal displacement $d\mathbf{r}$ is tangent to the path. The quantity

$$\mathbf{A} \cdot d\mathbf{r}$$

measures the component of $\mathbf{A}$ along the path, multiplied by the infinitesimal distance travelled. Its integral around the closed curve,

$$\oint_C \mathbf{A} \cdot d\mathbf{r},$$

is called the circulation of $\mathbf{A}$ around C . The curl provides a local measure of this tendency to circulate. In the infinitesimal limit,

$$(\nabla \times \mathbf{A}) \cdot \hat{n} = \lim_{S \rightarrow 0} \frac{1}{S} \oint_{\partial S} \mathbf{A} \cdot d\mathbf{r}.$$

The relation $\mathbf{B} = \nabla \times \mathbf{A}$ means that the magnetic field represents the local circulation density of the vector potential.

The local relation between curl and circulation can be extended to a finite surface through Stokes' theorem. Let S be an oriented surface whose boundary is the closed curve $C = \partial S$. Then

$$\oint_C A \cdot dr = \int_S (\nabla \times A) \cdot dS.$$

The orientation of C is related to the normal direction of S by the right-hand rule.

Using $B = \nabla \times A$,

$$\oint_C A \cdot dr = \int_S B \cdot dS.$$

The surface integral on the right-hand side is the magnetic flux through S ,

$$\Phi_B \equiv \int_S B \cdot dS = \oint_C A \cdot dr,$$

so the circulation of the vector potential around a closed curve is equal to the magnetic flux enclosed by that curve.

1.2.3 The electromagnetic potentials

In electrostatics, the electric field is obtained from the scalar potential through

$$E = -\nabla V.$$

This relation, however, is not sufficient when the magnetic field varies in time. In the general electromagnetic case, the electric and magnetic fields must be described together through the scalar potential V and the vector potential A .

Faraday's law states that

$$\nabla \times E = -\frac{\partial B}{\partial t}.$$

Using $B = \nabla \times A$, we obtain

$$\nabla \times E = -\frac{\partial}{\partial t}(\nabla \times A) = -\nabla \times \frac{\partial A}{\partial t}.$$

Therefore,

$$\nabla \times \left(E + \frac{\partial A}{\partial t} \right) = 0.$$

A curl-free vector field can locally be expressed as the gradient of a scalar function. Introducing the scalar potential V , we write

$$E + \frac{\partial A}{\partial t} = -\nabla V,$$

and hence

$$E = -\nabla V - \frac{\partial A}{\partial t}.$$

The electromagnetic fields are therefore related to the potentials by

$$E = -\nabla V - \frac{\partial A}{\partial t}, \quad B = \nabla \times A.$$

In the static limit, $\partial A / \partial t = 0$, and the familiar electrostatic relation $E = -\nabla V$ is recovered.

1.3 Gauge freedom

We have seen that the electromagnetic fields can be expressed in terms of the scalar and vector potentials as

$$E = -\nabla V - \frac{\partial A}{\partial t}, \quad B = \nabla \times A.$$

These relations determine the fields from the potentials, but they do not determine the potentials uniquely.

Consider a sufficiently regular scalar function

$$\chi(r, t).$$

From a given pair of potentials V and A , define a new pair by

$$A' = A + \nabla \chi, \quad V' = V - \frac{\partial \chi}{\partial t}.$$

Let us calculate the magnetic field generated by the transformed vector potential:

$$B' = \nabla \times A' = \nabla \times (A + \nabla \chi).$$

Using the identity $\nabla \times (\nabla \chi) = 0$ we obtain

$$B' = \nabla \times A = B.$$

The electric field is also unchanged:

$$E' = -\nabla V' - \frac{\partial A'}{\partial t} = -\nabla \left(V - \frac{\partial \chi}{\partial t} \right) - \frac{\partial}{\partial t} (A + \nabla \chi) = -\nabla V + \nabla \frac{\partial \chi}{\partial t} - \frac{\partial A}{\partial t} - \frac{\partial}{\partial t} \nabla \chi.$$

Since spatial and temporal derivatives commute,

$$\nabla \frac{\partial \chi}{\partial t} = \frac{\partial}{\partial t} \nabla \chi,$$

the two additional terms cancel, leaving

$$E' = -\nabla V - \frac{\partial A}{\partial t} = E.$$

The transformation

$$A' = A + \nabla \chi, \quad V' = V - \frac{\partial \chi}{\partial t}$$

is called a *gauge transformation*. Since the function $\chi(r, t)$ can be chosen arbitrarily without changing the electric and magnetic fields, the potentials are said to possess *gauge freedom*.

This generalizes the freedom already encountered for an ordinary scalar potential. Adding a constant to a potential does not change its gradient; similarly, adding the gradient of a scalar function to the vector potential does not change its curl. In the electromagnetic case, however, a time-dependent change in A must be accompanied by a corresponding change in V so that the electric field also remains unchanged.

Therefore the values of V and A at an individual point are not uniquely determined by the electromagnetic fields. Different pairs of potentials connected by a gauge transformation represent the same electric and magnetic fields and hence the same classical electromagnetic situation. Some quantities

constructed from potentials are nevertheless unchanged by a gauge transformation. For example, consider the circulation of the vector potential around a closed curve C :

$$\oint_C A' \cdot dr = \oint_C A \cdot dr + \oint_C \nabla \chi \cdot dr.$$

For a single-valued function χ ,

$$\oint_C \nabla \chi \cdot dr = 0,$$

and therefore

$$\oint_C A' \cdot dr = \oint_C A \cdot dr.$$

Using Stokes' theorem, this gauge-invariant circulation is equal to the magnetic flux enclosed by the curve:

$$\oint_C A \cdot dr = \int_S B \cdot dS = \Phi_B.$$

In classical electromagnetism, the measurable force on a charged particle depends only on E and B , so the non-uniqueness of the potentials may appear to make them just auxiliary mathematical objects.

1.4 Electromagnetic potentials in quantum mechanics

Up to this point nothing has forced us to take the potentials seriously. The classical equations of motion contain only E and B , and the gauge freedom

$$A \mapsto A + \nabla\chi, \quad V \mapsto V - \frac{\partial\chi}{\partial t},$$

with $\chi(r, t)$ an arbitrary function, never reaches an observable. Quantum mechanics removes this comfort.

1.4.1 Mathematical interlude II: the quantum Hamiltonian

In classical mechanics, the motion of a particle is determined by the force acting on it: Newton's second law fixes the acceleration, and the trajectory follows by integration. Quantum mechanics is formulated differently. The state of the particle is described by a wavefunction $\psi(r, t)$, whose time evolution is governed by an operator called the Hamiltonian,

$$i\hbar \frac{\partial\psi}{\partial t} = \hat{H}\psi. \quad (1.1)$$

For the non-relativistic system considered here, the Hamiltonian represents the total energy of the particle.

The Hamiltonian is the operator associated with the total energy of the system. Equation (1.1) states that the energy operator *generates* the evolution: the change of ψ during an infinitesimal interval dt is prescribed by $\hat{H}\psi$. To specify the dynamics of a quantum system is therefore to specify its energy, in the same sense in which specifying the dynamics of a classical particle means specifying the force acting on it.

Classical mechanics admits an equivalent formulation, due to Hamilton, in which the evolution is generated by the function

$$H(r, p) = \frac{p^2}{2m} + U(r), \quad \dot{r} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial r},$$

rather than by F . The trajectories obtained are exactly the Newtonian ones; what changes is the object taken as primitive. Hamilton's formulation is written in terms of the potential energy, rather than Newton's, which is written in terms of forces. In quantum mechanics this choice becomes a choice of what the theory regards as fundamental.

The simplest consequence of eq. (1.1) fixes the meaning of the Hamiltonian. If U does not depend on time, there exist solutions of definite energy,

$$\psi(r, t) = \psi_E(r) e^{-iEt/\hbar},$$

for which the probability density $|\psi|^2 = |\psi_E|^2$ is independent of time. The time evolution of a stationary state consists entirely of a phase factor. This is the first indication of a structural feature of quantum mechanics: a global phase, $\psi \mapsto e^{i\alpha}\psi$ with α constant, leaves every prediction unchanged and is therefore not an observable. A *relative* phase between two components of a superposition is a different matter.

If $\psi = \psi_1 + \psi_2$ then

$$|\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\Re(\psi_1^*\psi_2)$$

and the interference term depends on the phase difference between the two branches. Quantum mechanics thus provides an observable that has no classical counterpart, and it is through this

observable that the electromagnetic potentials will enter the physics.

1.5 Minimal coupling

A quick recap of the basics of quantum mechanics is in order before proceeding.

In quantum mechanics the state of a particle is a function $\psi : \mathbb{R}^3 \rightarrow \mathbb{C}$. At every point in space we have a complex number; this can be visualized as a bizarre clock. The modulus $|\psi|$, the length of the hand, is not itself a probability: its square $|\psi|^2$ gives the probability density of finding the particle in that precise spot in space; the argument tells us where the hand is pointing.

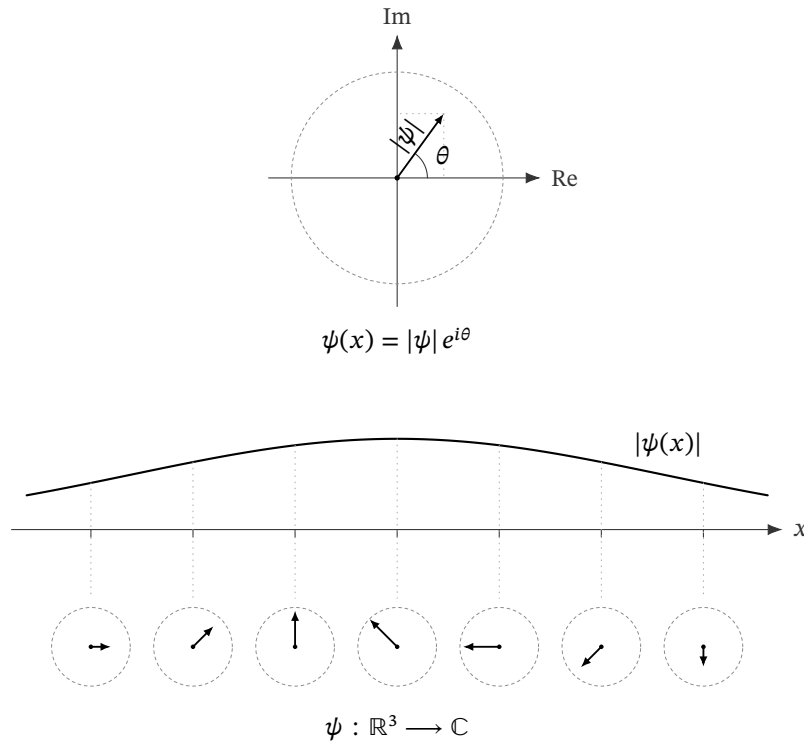


Figure 1.4. Clock representation of a complex wavefunction. At each position, the length of the hand gives the modulus $|\psi(x)|$, while its angle gives the phase of $\psi(x)$.

The modulus is measurable, the direction of the hand is not. If we rotate every hand by the same angle, $\psi \mapsto e^{i\alpha}\psi$, we do not change any measurement. The absolute phase is just a convention, like choosing the prime meridian.

What is measurable is a difference in phase. If two amplitudes contribute at the same point, $\psi_1 + \psi_2$, the squared modulus contains the cross term $2\Re(\psi_1^*\psi_2)$, which depends on the relative angle between the two hands. When they are aligned, they add; when they point in opposite directions, they cancel. This is the only way in which phase becomes observable, and it's the entire mechanism behind interference.

1.5.1 Momentum is the hand's rotation rate

Take the plane wave $\psi(x) = e^{ikx}$. Its magnitude is constant, but the hand rotates while advancing in space, at a rate k : k radians of phase per unit length. The faster it rotates, the shorter the wavelength. This concept is explained by the de Broglie relation; to a short wavelength corresponds a large momentum

$$p = \hbar k.$$

And we can measure the spatial rate at which the phase rotates by differentiating:

$$-i\hbar \frac{\partial}{\partial x} e^{ikx} = \hbar k e^{ikx} = p e^{ikx}.$$

1.5.2 Comparing phases at different points

When we write $\nabla\psi$ we are comparing the value of the wavefunction at x with its value at a nearby point $x + dx$. However, such a comparison requires a convention that defines when they are to be regarded as equal.

Since the absolute phase of a wavefunction has no physical meaning, there is no unique, physically preferred phase convention throughout space. The ordinary gradient ∇ implicitly assumes that the same phase convention is being used at every point. This is a choice of gauge, not an observable fact.

The vector potential A provides the correction needed to compare phases defined in different local conventions. Over an infinitesimal displacement dx , the corresponding phase correction is

$$\frac{q}{\hbar} A \cdot dx.$$

The ordinary gradient therefore contains both the physical variation of the state and a contribution arising from the chosen local phase convention. To isolate the gauge-covariant variation, we introduce the covariant derivative

$$D = \nabla - \frac{iq}{\hbar} A.$$

Multiplying by $-i\hbar$ we obtain

$$-i\hbar D = -i\hbar \nabla - qA.$$

This is minimal coupling: the ordinary derivative is replaced by a covariant derivative that compares the wavefunction consistently across different points.

The operator

$$\hat{p} = -i\hbar \nabla$$

represents the canonical momentum, whereas

$$\hat{\pi} = \hat{p} - qA = -i\hbar \nabla - qA$$

is the kinetic momentum, related to the particle's velocity by $\hat{\pi} = m\hat{v}$. So in the presence of a vector potential:

$$\hat{\pi} = m\hat{v}, \quad \hat{p} \neq \hat{\pi}.$$

If the argument of ψ at a point is only defined relative to some reference, then nothing stops us from choosing that reference differently at every point. We can pick any function $\chi(x)$ and rotate the phase at x by the angle $\frac{q}{\hbar}\chi(x)$:

$$\psi \longmapsto e^{iq\chi/\hbar} \psi.$$

This is a relabelling, not a change of state. The modulus is untouched, $|e^{iq\chi/\hbar}\psi|^2 = |\psi|^2$, so every

probability is exactly what it was.

We now recall that A recorded how much the reference rotates when we step from x to $x + dx$. When re-defining the reference point by point, we have also changed the rate at which it rotates between neighbouring points, and that rate changes precisely by the added gradient

$$A \longmapsto A + \nabla\chi.$$

The two modifications are not independent. One is forced by the other, and together they cancel exactly. Applying ∇ to the rotated wavefunction produces an extra term from differentiating the exponential,

$$\nabla (e^{iq\chi/\hbar}\psi) = e^{iq\chi/\hbar} \left(\nabla\psi + \frac{iq}{\hbar}(\nabla\chi)\psi \right),$$

and the shifted potential $A + \nabla\chi$ subtracts exactly that extra term back out. The combination $-i\hbar\nabla - qA$ therefore survives unchanged, up to the overall phase factor. No measurement notices anything.

1.6 The Schrödinger equation

We now have every element needed to write the equation of motion of a charged particle. The quantum Hamiltonian fixes the dynamics of a quantum system; minimal coupling tells us that, in the presence of an electromagnetic field, the operator $-i\hbar\nabla$ no longer represents the momentum associated with the motion of the particle. Putting the two statements together produces the Schrödinger equation for a particle of charge q in an external electromagnetic field.

For a free particle the classical Hamiltonian is the kinetic energy alone,

$$H = \frac{p^2}{2m},$$

and the canonical quantization prescription $p \mapsto \hat{p} = -i\hbar\nabla$ gives

$$\hat{H}_0 = -\frac{\hbar^2}{2m}\nabla^2.$$

The kinetic energy is always $\frac{1}{2}mv^2$. What changes in the presence of a field is which operator represents mv : that role belongs to the kinetic momentum

$$\hat{\pi} = \hat{p} - qA = -i\hbar\nabla - qA,$$

and not to the canonical momentum $\hat{p}$. The kinetic term must therefore be built from $\hat{\pi}$,

$$\hat{T} = \frac{\hat{\pi}^2}{2m} = \frac{1}{2m}(-i\hbar\nabla - qA)^2.$$

The scalar potential enters as in the classical case: a charge q placed at position r has electrostatic potential energy $U = qV(r)$, which acts on the wavefunction by multiplication. The Hamiltonian of a charged particle in an electromagnetic field is therefore

$$\hat{H} = \frac{1}{2m}(-i\hbar\nabla - qA)^2 + qV,$$

and the Schrödinger equation reads

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi = \left[\frac{1}{2m}(-i\hbar\nabla - qA)^2 + qV \right] \psi.$$

Expanding the square requires some care, because ∇ and A do not commute; the operator ∇ acts on both A and ψ . Applying $-i\hbar\nabla - qA$ twice,

$$(-i\hbar\nabla - qA)^2\psi = -\hbar^2\nabla^2\psi + i\hbar q(\nabla \cdot A)\psi + 2i\hbar qA \cdot \nabla\psi + q^2A^2\psi,$$

so that

$$i\hbar\frac{\partial\psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + \frac{i\hbar q}{2m}(\nabla \cdot A) + \frac{i\hbar q}{m}A \cdot \nabla + \frac{q^2A^2}{2m} + qV \right] \psi.$$

The terms linear in A are the ones responsible for the effects we are interested in. The term $q^2A^2/2m$ is quadratic in the potential and in the charge, not in the field, and it is not negligible in general: it is responsible, for instance, for the diamagnetic response of atoms. Nor does the argument that follows need to drop it: the Aharonov–Bohm phase is derived from the full operator $(-i\hbar\nabla - qA)^2$ and is exact. If the gauge is chosen so that $\nabla \cdot A = 0$, the two first-order terms collapse into a single one.

1.6.1 Gauge covariance

The equation just written contains the potentials, not the field, and the potentials are not unique. We must therefore check that the physics it describes does not depend on that choice.

Perform a gauge transformation and, at the same time, the local phase rotation introduced in the previous section:

$$A' = A + \nabla\chi, \quad V' = V - \frac{\partial\chi}{\partial t}, \quad \psi' = e^{iq\chi/\hbar} \psi.$$

The key property is the one already established for the covariant derivative: the extra term produced by differentiating the exponential is exactly cancelled by the extra term in A' ,

$$(-i\hbar\nabla - qA')\psi' = e^{iq\chi/\hbar}(-i\hbar\nabla - qA)\psi,$$

and, applying the same identity twice,

$$(-i\hbar\nabla - qA')^2\psi' = e^{iq\chi/\hbar}(-i\hbar\nabla - qA)^2\psi.$$

The time derivative produces an extra term of its own,

$$i\hbar\frac{\partial\psi'}{\partial t} = e^{iq\chi/\hbar}\left(i\hbar\frac{\partial\psi}{\partial t} - q\frac{\partial\chi}{\partial t}\psi\right),$$

while the transformed scalar potential contributes

$$qV'\psi' = e^{iq\chi/\hbar}\left(qV - q\frac{\partial\chi}{\partial t}\psi\right).$$

The same term $-q(\partial\chi/\partial t)\psi$ appears on both sides and cancels, leaving

$$i\hbar\frac{\partial\psi}{\partial t} = \left[\frac{1}{2m}(-i\hbar\nabla - qA)^2 + qV\right]\psi,$$

that is, the original equation. The Schrödinger equation is therefore *covariant* under gauge transformations: it does not leave ψ unchanged, but it retains its form, and since $|\psi'|^2 = |\psi|^2$ no prediction is affected.

This is a stronger statement than the classical one. Classically, gauge freedom was a redundancy in the description of E and B ; here it is inseparably tied to the freedom of choosing the phase convention of the wavefunction at each point. Fixing a gauge for A and fixing a local phase convention for ψ are the same act, and neither of the two changes can be made without the other.

The symmetry between the scalar and the vector potential becomes explicit if we introduce the covariant derivative in time as well,

$$D_t = \frac{\partial}{\partial t} + \frac{iq}{\hbar}V, \quad D = \nabla - \frac{iq}{\hbar}A,$$

so that the equation takes the compact form

$$i\hbar D_t\psi = \frac{1}{2m}(-i\hbar D)^2\psi.$$

The scalar and vector potentials enter through a single prescription, applied to derivatives in time and in space alike.

Chapter 2

The Aharonov–Bohm Effect

The classical dismissal of the potentials rests on two claims.¹ First, the equations of motion contain only E and B . Second, the potentials carry *more* information than the fields, and this excess is pure gauge, hence unphysical. Aharonov and Bohm’s starting point is to examine the second claim rather than accept it [1]. The same interference shift had already been anticipated by Ehrenberg and Siday in the context of electron optics [5].

2.1 The loss of information

The relation

$$B = \nabla \times A$$

is a differentiation. Like every differentiation, it destroys information: the map $A \mapsto B$ is not injective. The standard argument is that everything destroyed is gauge, that is, nothing. It is worth checking whether this is true.

Suppose B is known throughout a region $\mathcal{R}$, and let A and A' be two vector potentials reproducing it. Then

$$\nabla \times A' = \nabla \times A \implies \nabla \times (A' - A) = 0,$$

so $A' - A$ is curl-free, the curl is 0. One would like a curl-free field to be the gradient of a single-valued scalar function, but this is not automatic; it has to be built. Define

$$\chi(r) = \int_{r_0}^r (A' - A) \cdot dr'.$$

For this to define a function at all, the integral must not depend on which path we take from r_0 to r . Take two such paths: together they form a closed loop C , and path-independence is the same as asking

$$\oint_C (A' - A) \cdot dr = 0.$$

By Stokes’ theorem this integral equals the flux of the curl through any surface bounded by C , and the curl vanishes. So the answer is zero, and χ is well defined. The catch is in the “any surface bounded by C ”: that surface must lie inside $\mathcal{R}$. If $\mathcal{R}$ is simply connected (a region is simply connected if every closed loop inside it can be shrunk continuously to a point without ever leaving the region) it always does, and the argument goes through. If $\mathcal{R}$ has a hole and C encircles it, every capping surface has to cross the hole, i.e. leave $\mathcal{R}$, where we know nothing about the curl. Stokes no longer applies, and the conclusion genuinely fails.

So on a multiply connected region, knowing B does not determine A up to a gauge transformation. What survives the differentiation is a gauge-invariant quantity that we have already met, the circulation

$$\oint_C A \cdot dr = \Phi_B,$$

¹For an accessible popular introduction to the effect, including the toroidal experiment of section 2.4, see [10].

one number for each class of non-contractible loops; two loops that wind around the hole the same number of times give the same integral. The classes are labelled by $n \in \mathbb{Z}$, and the value is $n\Phi_B$, so a single real parameter generates everything.

This quantity is not attached to a point but to a loop: it is non-local by construction; and it may be nonzero on a curve along which B vanishes identically.

This is the sense in which replacing the potential by the field means accepting a loss of information. Classical mechanics never registers the loss, because the Lorentz force is evaluated pointwise, where the field is; a particle confined to a region with $B = 0$ feels no force regardless of Φ_B . The information is discarded because nothing in the classical theory can read it.

One objection must be addressed immediately, since it is the crux of the argument. The quantity Φ_B is itself constructed from the field: by Stokes' theorem it is the flux of B through any surface bounded by C . One could therefore insist that no new entity is needed. Aharonov and Bohm accept the algebra but reject the interpretation: the flux resides in a region the electron never enters, and in a theory of local interaction a field cannot act where it is absent. Either the field acts non-locally, or the potentials are physically effective in the field-free region. The effect does not decide which language is preferable; it shows that the field description alone, taken locally, is incomplete.

Quantum mechanics reads the discarded information because it couples to A along a path. With the minimal coupling Hamiltonian

$$\hat{H} = \frac{1}{2m}(\hat{p} - qA)^2 + qV,$$

a solution in a field-free region can be written as the field-free solution multiplied by a phase factor,

$$\psi(r) = \psi_0(r) \exp \left[\frac{iq}{\hbar} \int^r A \cdot dr' \right].$$

If the region is simply connected the integral depends only on the endpoint, the phase is single-valued, and the potential has been removed by a gauge transformation. If the region encircles a hole the integral depends on the path, and the phases accumulated along two paths on opposite sides of the hole differ by $\Delta\delta = \delta_1 - \delta_2$. Taking C along the first path and back along the second gives

$$\Delta\delta = \frac{q}{\hbar} \oint_C A \cdot dr = \frac{q\Phi_B}{\hbar}.$$

The loss of information reappears as a relative phase.

2.2 The time-dependent scalar arrangement

The arrangement proposed in the original paper isolates exactly this situation. Consider a charged particle inside a closed conductor connected to an external generator that makes the potential of the cage vary in time. Inside a conductor the field vanishes, so the particle experiences $E = 0$ throughout; yet the potential $V(t)$ is nonzero, and it adds to the Hamiltonian a term

$$qV(t),$$

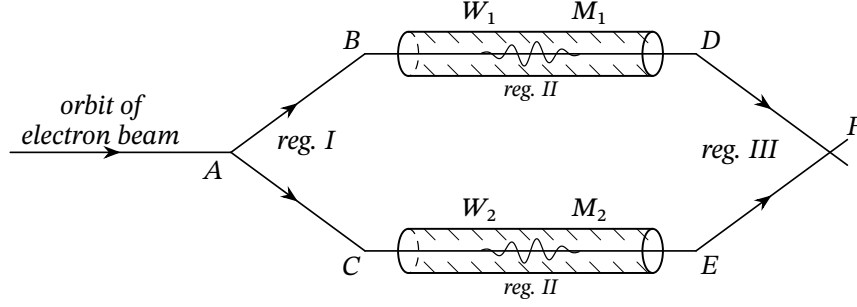


Figure 2.1. Aharonov and Bohm's proposed arrangement for the time-dependent scalar-potential effect. The electron beam is split at A; the two wave packets W_1 and W_2 traverse the conducting tubes M_1 and M_2 , then recombine in the interference region at F. Redrawn after Aharonov and Bohm [1, Fig. 1].

which inside the cage is a function of time only. Writing $\hat{H} = \hat{H}_0 + qV(t)$, where $\hat{H}_0$ is the Hamiltonian with the generator switched off, one verifies directly that if ψ_0 solves the problem for $\hat{H}_0$ then

$$\psi = \psi_0 e^{-iS/\hbar}, \quad S = q \int V(t) dt$$

solves the problem. Indeed

$$i\hbar \frac{\partial \psi}{\partial t} = \left(i\hbar \frac{\partial \psi_0}{\partial t} + \psi_0 \frac{\partial S}{\partial t} \right) e^{-iS/\hbar} = (\hat{H}_0 + qV(t))\psi,$$

where the last step uses $\partial S / \partial t = qV(t)$ and the fact that S does not depend on position, so that the phase factor commutes with the kinetic term. This spatial uniformity is the whole point: $\nabla V = 0$ means no force, and it is precisely what allows the potential to act as a pure, position-independent phase. Taken by itself the result is unremarkable: a single wavefunction multiplied by a global phase describes the same physical state.

The two-tube geometry. The arrangement of fig. 2.1 turns that phase into an observable one by making it *relative*. A coherent electron beam is split at A into two parts, which travel through the two branches ABD and ACE and are recombined at F. Each branch contains a long cylindrical metal tube, M_1 and M_2 , each acting as a Faraday cage held at its own potential $V_1(t)$, $V_2(t)$.

Electrical shutters chop the beam into wave packets W_1 and W_2 , long compared with the electron wavelength λ but short compared with the length of the tubes. A time-delay mechanism then fixes the potentials so that:

- in region I, before each packet enters, both potentials are zero;
- in region II, once each packet is well inside its own tube, the potentials grow, differently in the two tubes;

- they fall back to zero before the packet approaches the far edge, so that in region III there is again no potential.

The timing is what makes the arrangement work. The electric field associated with a changing tube potential does not penetrate far beyond the ends of the tube, and it is nonzero only at times when the packet is deep inside, far from the ends. The electron is therefore exposed to a time-varying potential without ever entering a region where the field is present.

The relative phase. Each packet acquires the phase computed above, and the recombined wave-function is

$$\psi = \psi_1^0 e^{-iS_1/\hbar} + \psi_2^0 e^{-iS_2/\hbar}, \quad S_i = q \int V_i dt.$$

The interference at F depends on the difference

$$\Delta\delta = \frac{S_2 - S_1}{\hbar} = -\frac{q}{\hbar} \int (V_1 - V_2) dt,$$

which is nonzero whenever the two tubes are driven differently. The pattern is displaced although no force was exerted on either packet at any moment of its history.

The space-time reading. The phase difference can be written as a circulation around the closed circuit formed by the two branches, with the same orientation,

$$\Delta\delta = -\frac{q}{\hbar} \oint_C V dt,$$

closed not in space but in space-time. Relativistic covariance then demands that the time component cannot appear alone: the invariant object is

$$\Delta\delta = \frac{q}{\hbar} \oint_C (A \cdot dr - V dt),$$

integrated over any closed circuit in space-time. The arrangement of fig. 2.1 realises the scalar-potential term. The specialisation to a purely spatial circuit, $t = \text{constant}$, leaves the vector-potential term, and yields the phase difference already obtained in section 2.1,

$$\Delta\delta = \frac{q}{\hbar} \oint_C A \cdot dr = \frac{q\Phi_B}{\hbar}.$$

The two arrangements are therefore one argument. In both, the electron worldlines bound a region of space-time through which the electromagnetic field has non-vanishing flux, while the field vanishes on the worldlines themselves.

2.3 The magnetic Aharonov–Bohm arrangement

The second arrangement proposed in the original paper isolates exactly the situation of section 2.1. A coherent electron beam is split into two parts which pass on opposite sides of a long, tightly wound solenoid and are recombined in an interference region.

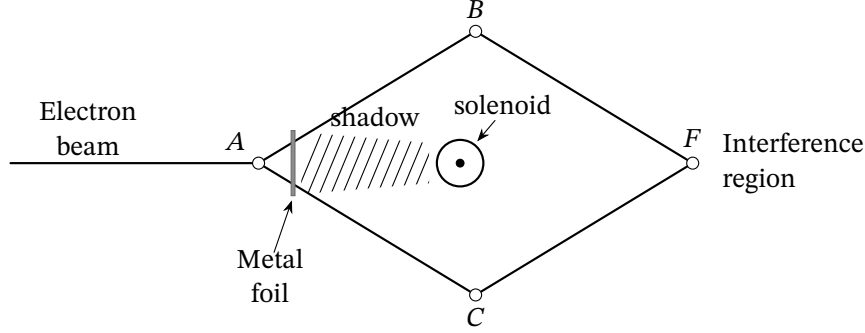


Figure 2.2. Aharonov and Bohm’s proposed arrangement for the magnetic effect. A metal foil splits the electron beam into two paths, which pass on opposite sides of the solenoid and recombine in the interference region at F . The magnetic field is confined to the solenoid, while the two paths enclose its magnetic flux. Schematic based on Aharonov and Bohm [1].

The solenoid confines the magnetic field to its interior; outside $B \approx 0$, while A cannot vanish identically, since every loop enclosing the axis must return the same flux. To exclude any contact between the electrons and the field region, the solenoid is screened from the beam, in the original proposal by a thin plate casting a shadow. Each partial beam travels in a simply connected region, so the phase along it is well defined; the two regions together enclose the flux. The relative phase at recombination is

$$\Delta\delta = \frac{q\Phi_B}{\hbar} = -\frac{e\Phi_B}{\hbar}$$

for an electron of charge $q = -e$. The observable consequence is a rigid displacement of the interference pattern, with the fringe spacing unchanged, periodic in the enclosed flux with period

$$\Phi_0 = \frac{h}{e} \approx 4.14 \times 10^{-15} \text{ Wb}.$$

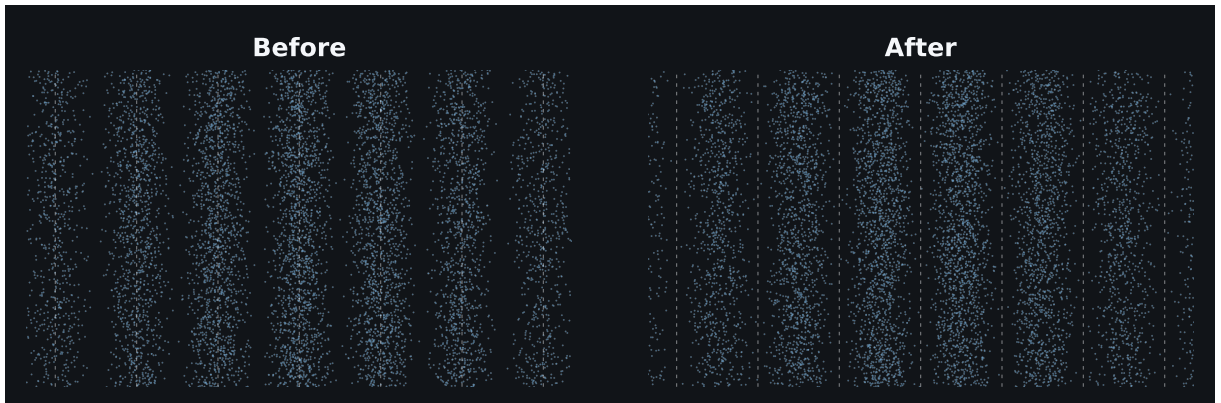


Figure 2.3. Author-generated Monte Carlo illustration of single-electron accumulation before and after inserting half a flux quantum, $\Phi_B = \Phi_0/2$. The resulting Aharonov–Bohm phase shift $\Delta\delta = -\pi \equiv \pi \pmod{2\pi}$ translates the fringes by half a period without changing the diffraction envelope. The dashed lines mark the original maxima.

No force is exerted on the electrons at any point of their trajectories.

Aharonov and Bohm already observed that comparing patterns recorded with and without flux is

experimentally awkward, and suggested instead varying the flux within a single exposure, so that the alteration of the fringes is recorded directly.

2.4 The toroidal experiment of Tonomura

The prediction of a fringe shift controlled by an inaccessible flux was tested soon after the original paper, first by Chambers [4] and later, with a biprism-and-solenoid geometry, by Möllenstedt and Bayh [6]. These experiments observed the expected displacement of the interference pattern, but they were open to a persistent objection: no real solenoid is infinite, and a finite solenoid leaks a small stray field into the region crossed by the electrons. A sceptic could therefore attribute the observed shift not to the vector potential in a field-free region, but to the electrons interacting with this residual magnetic field along their paths.

The experiment that closed the loophole was performed by Tonomura and collaborators in 1986 [9]. The idea is to replace the solenoid with a geometry that has no ends from which the field can leak, a toroid (a ring). The sample is a microscopic toroidal magnet of permalloy, a few micrometres across, in which the magnetic flux circulates entirely inside the ring. An electron wave is then split so that one part passes through the hole of the toroid and the other part passes outside it. The two partial waves enclose the magnetic flux Φ_B trapped in the ring, so their relative phase at recombination is

$$\Delta\delta = \frac{q}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{r} = \frac{q\Phi_B}{\hbar},$$

exactly as in the solenoid case, while neither wave ever travels through a region of non-zero magnetic field.

Two further precautions make the shielding airtight. First, the toroid is covered with a layer of niobium, a superconductor. Below its critical temperature, a superconductor expels magnetic fields from its interior (the Meissner effect), so the superconducting layer acts as a wall that the trapped flux cannot cross: any residual leakage field is suppressed, not merely assumed small. Second, the whole structure is coated with copper, which prevents the electron wave from penetrating into the magnet itself. The electrons therefore travel only through regions in which $B = 0$, and the flux is confined by a physical mechanism rather than by an idealization.

The superconducting layer has a second consequence, which turns a precaution into a signature. The magnetic flux trapped inside a superconducting ring cannot take arbitrary values: it is quantized in integer multiples of the superconducting flux quantum $h/2e$.² For an electron of charge $q = -e$, setting $\Phi_B = n h/2e$, with $n \in \mathbb{Z}$, in the phase difference gives

$$\Delta\delta = -\frac{e}{\hbar} n \frac{h}{2e} = -n\pi.$$

The predicted relative phase between the wave through the hole and the wave outside is therefore, modulo 2π , either 0 (for even n) or π (for odd n), and nothing in between. A phase of π corresponds to a displacement of the interference fringes inside the hole by exactly half a fringe spacing with respect to the fringes outside.

This is precisely what was observed. When the sample is cooled below the critical temperature of niobium and imaged by electron interferometry, the fringes crossing the hole of the toroid appear

²The elementary flux quantum associated with a charge q is $\Phi_0 = h/|q|$. In a superconductor the charge carriers are Cooper pairs of charge $2e$, so the trapped flux is quantized in units of $h/2e$, half the value one would associate with a single electron.

shifted by half a spacing relative to those outside, for toroids trapping an odd number of flux quanta, and unshifted for an even number. The interference micrographs show this directly: the fringe system inside the hole is displaced by half a period with respect to the reference fringes, even though the magnetic field is confined inside the shielded ring.

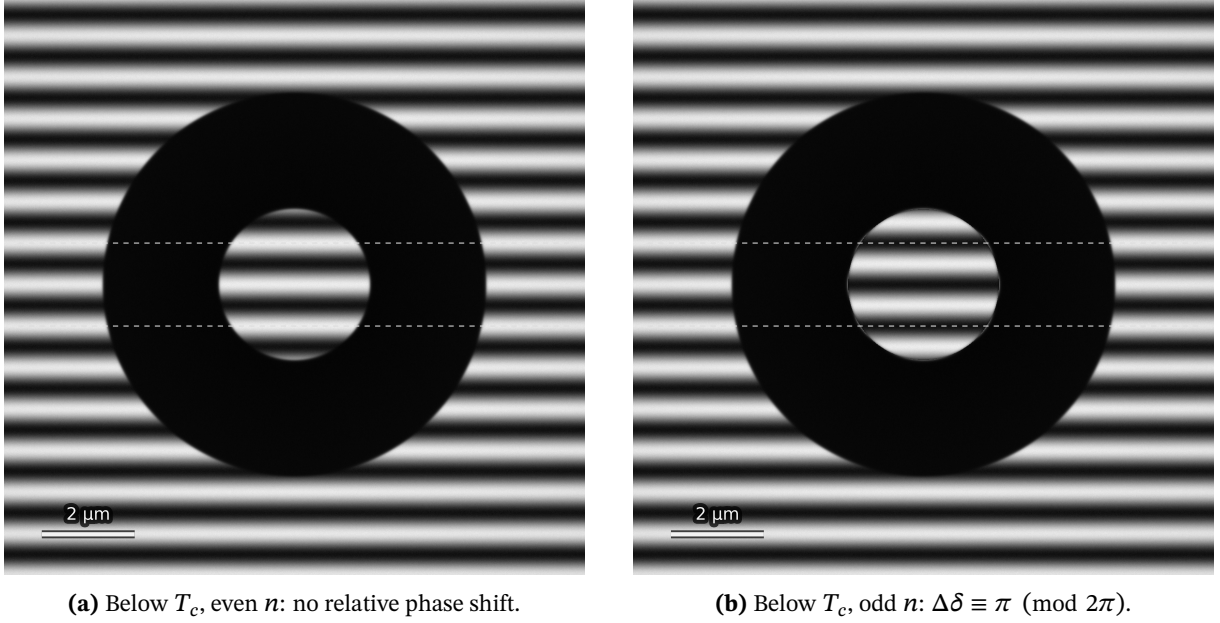


Figure 2.4. Schematic comparison of even and odd trapped flux quanta below T_c . The fixed dashed guides mark an outside maximum: with no phase offset the same maximum continues through the hole, whereas the Aharonov–Bohm phase moves the interior pattern by half a fringe spacing. Author-generated schematic based on the toroidal experiment of Tonomura et al. [9].

The result leaves no room for the stray-field objection. The electrons move exclusively in a region where $B = 0$, enforced by the Meissner effect rather than by geometry alone, and yet their interference pattern records the enclosed flux. What the electrons respond to is not the field along their paths, which vanishes, but the gauge-invariant circulation $\oint_C A \cdot dr = \Phi_B$, which does not. The Aharonov–Bohm phase is therefore a genuine physical effect: a non-local, gauge-invariant functional of the vector potential is observable even when the field itself is everywhere inaccessible.

2.5 Concluding remarks

In classical physics the potentials can be regarded as auxiliary objects, since the force on a charge depends only on E and B . In quantum mechanics this is no longer the case. The Schrödinger equation of a charged particle is written in terms of V and A through minimal coupling, and the fields never appear in it explicitly. Gauge freedom survives, but only as the joint freedom of redefining the potentials and the local phase convention of the wavefunction.

What the potentials make observable is not their value at a point, which is gauge-dependent, but the phase factor

$$\exp \left[\frac{iq}{\hbar} \oint_C A \cdot dr \right] = \exp \left[\frac{iq\Phi_B}{\hbar} \right],$$

fixed by the flux linked by the loop C . When $B = 0$ throughout the region accessible to the particle, this factor is unity in a simply connected region, where the vector potential can be removed by a gauge transformation; around a hole it can differ from unity, depending on the enclosed flux.

Wu and Yang [11] later turned this observation into a general principle: the field strength alone underdescribes electromagnetism, and the complete gauge-invariant description is given by the phase factors around closed loops.

The prediction has been confirmed experimentally. After the early observations of Chambers [4] and of Möllenstedt and Bayh [6], the toroidal experiment of Tonomura and collaborators removed the stray-field objection by confining the flux with a superconducting layer, and observed relative phases of 0 or π , as required by flux quantization [9].

The phase discussed here is also the first example of a broader class. Berry [3] showed that a quantum system transported adiabatically around a closed loop in parameter space acquires a geometric phase, and treated the Aharonov–Bohm phase as a particular case. Aharonov and Casher [2] showed that a neutral particle with a magnetic moment, moving around a line of charge, acquires an analogous topological phase, the dual of the magnetic arrangement discussed above. Comprehensive reviews of the theory and of the experiments can be found in [7, 8].

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